

Yuqian Zhang

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EDUCATION

Duke Kunshan University (DKU) & Duke University Dual Degree	08/2022-05/2026
B.S. in Applied Math & Computational Science; Math Track (DKU)	Kunshan, China
B.S. in Interdisciplinary Studies: Applied Math & Computational Science; Math (Duke)	Durham, U.S.
- GPA: 3.85/4.0; Dean's List Fall 2023, Spring 2024, Spring-Fall 2025 (w. Distinction) Latin Honors – magna cum laude - Courses: Stochastic Differential Equation, Measure Theory, Numerical Analysis, Advance Linear Algebra, Algorithm Analysis & Design, Graphics Software Architecture, Image Data Science.	

AWARDS & SCHOLARSHIPS

DKU Student Experiential Learning Fellowship Program	2025 Summer Kunshan, China
Duke Undergraduate Semester Research Program Reward	2025 Spring Durham, U.S.
DKU Summer Research Scholars	2024 Summer Kunshan, China

RESEARCH EXPERIENCE

Spatial Transcriptomics Data Denoising through GVAE and Mass Conservation	05/2025-Now
Research Intern with Dr. Yi Zhang, Professor of Neurosurgery	Durham, U.S.
- Developed a light-weight Graph Variational Auto-Encoder (GVAE) model to denoise spatial transcriptomics data by addressing zero-inflated observation noise and gene diffusion, ensuring localized graph structure and mass conservation while reconstructing a zero-inflated negative binomial distribution. - Incorporated probabilistic programming frameworks (Pyro) and recent GVAE techniques for robust modeling and inference. Achieved accurate reconstruction of gene expression fields from sparse samples and improved downstream tasks such as latent clustering and cell/bin-level classification. Gained deeper understanding of spatial dynamics and fluid-like diffusion processes in biological data. - Achieved enhancement of downstream tasks such as latent clustering, cell-level and bin-level classification. Work will be published this year.	
Enhancing Ocean Surface Oxygen Representation via Relative Spatial Embedding	08/2024-Now
Research Assistant with Dr. Zuchuan Li, Professor of Data and Environmental Sciences	Kunshan, China
- Investigate the ability of relative spatial embedding in enhancing ocean surface feature representation. Reproduced, modified, and examined several models, including Vision Transformer, Positional-induced Transformer, and Relative Position Embedding. Implement a satellite data matching system and corresponding data loader. - Gained fundamental knowledge in Transformers, spatial embedding, and representation learning. Processed massive time-series data with hundreds of gigabytes.	
Gene Expression Field Estimation via Poisson Resampling and Neural ODEs	12/2024-05/2025
Research Intern with Dr. Yi Zhang, Professor of Neurosurgery	Durham, U.S.
- Formulated Spatial Transcriptomics (ST) data as a spatial Poisson sampling process on a diffused gene expression field. Proposed a deep learning algorithm that generates reliable Poisson samples through a reversed Neural ODE process. Proposed several techniques, including a combination of supervised pretraining and semi-supervised operator learning. - Gained a deep understanding of fluid dynamics simulation. Achieved the accurate reconstruction of the mean field from sparse samples.	
Solving Time Series Inverse Problem by Combining Bayesian-PINN with Kalman Filter	11/2023-11/2024
Research Assistant with Dr. Zuchuan Li, Professor of Data and Environmental Sciences	Kunshan, China
Proposed a deep learning model in solving the inverse problem of field dynamics systems, acquired a deep understanding of Bayesian neural networks (BNNs), physics-informed neural networks (PINNs), and stochastic differential equations (SDEs). Proposed several techniques, including the patched covariance method, to improve Kalman filter performance.	

- Successfully implemented B-PINN-Kalman Filter in Python (Pytorch), replicated several papers about flow estimation, image generation, and time series inverse problem. Implemented a 2D advection-diffusion fluid solver with a CUDA acceleration as a neural operator.

General Study of Denoising Diffusion Probability Model

11/2023-05/2024

Independent Study with Dr. Zuchuan Li, Professor of Data and Environmental Sciences

Kunshan, China

- Independently studied the score-based diffusion model on the General Inverse Problem. Conducted an in-depth study of stochastic differential equations, score-based diffusion process, diffusion posterior sampling, and related mathematical background.
- Successfully implemented diffusion posterior sampling in Python (Pytorch), additionally replicated several papers about DDPM and diffusion sampling for both linear & nonlinear inverse problems.

PROFESSIONAL EXPERIENCE

WebGL Implementation of Reflective Gaussian Splatting (Ref-Gaussian)

08/2025-05/2026

Signature Work with Dr. Shixin Xu

Kunshan, China

- Conducted an in-depth survey of recent 3D Gaussian Splatting methods and identified performance and deployment limitations in existing browser-native implementations. Implemented the latest Ref-Gaussian pipeline in WebGL, achieving 60+ FPS real-time rendering with optimized loading and memory efficiency.
- Proposed Approximate Tangent Axis (ATA) estimation for Gaussian Surfels rasterization, producing smoother position maps and improved visual stability. Combined with early alpha discard, view-frustum culling, and chunk-based lazy updates, significantly improved rendering throughput and reduced GPU utilization, enabling real-time rendering on consumer-grade laptops.
- Built an end-to-end pipeline including COLMAP reconstruction, Ref-Gaussian inference, data post-processing, and Docker-based web deployment, enabling scalable and reproducible browser-based rendering.
- [XDzzzzZyq/dku-splat: WebGL implementation of Ref-Gaussian for Campus Navigation](#)

GPU(OpenGL) based Renderer Development

05/2022-Now

Independent Project

- Led long-term development of a real-time rendering engine for over 2 years, producing 15,000+ lines of C++/OpenGL code across 450+ commits. Studied 30+ foundational and state-of-the-art graphics papers, including reproductions of 3 SIGGRAPH works.
- Conducted in-depth exploration of physically based rendering, radiometry, high-quality shadowing, signed distance fields, spatio-temporal denoising, and real-time global illumination.
- Acquired strong expertise in GPU architecture, high-performance computing, and shader programming, with advanced proficiency in C++ and modern OpenGL for real-time graphics systems.
- [XDzzzzZyq/Opengl: OpenGL renderer development by XDzZyq / Documentation](#)

Blender Development

02/2024-Now

Individual Contributor; [Blender](#)

Remote

- Contributed to Blender's rendering and geometry editing pipeline by enhancing curve sculpting functionality, including curve radius initialization, interpolation schemes, and corresponding UI integration and technical documentation. Also improved features such material property editing and curve mapping UI interaction.
- Developed strong expertise in medium- to large-scale C++ graphics systems, gaining deep insight into 3D software architecture, rendering workflows, and collaborative Git-based development; 5 pull requests accepted in the official Blender repository.
- [Blender Constribution Reports - Blender Projects](#)

CUDA Image Processing Operator Development

9/2024-12/2024

Course Project

Kunshan, China

- Developed a CUDA-accelerated Python library for Image Processing (IP) and Computer Vision, achieving high-throughput GPU performance and scalable execution. Implemented core algorithms including convolution, geometric transformations, dithering, morphological operations, and texture segmentation.
- Gained in-depth expertise in CUDA programming and high-performance computing, and explored hybrid pipelines that integrate Pytorch, CUDA, and C++ for efficient parallel computing and research prototyping.
- [XDzzzzZyq/cuda-vision: CS 207 Homeworks](#)

Humanities Research Center at DKU

12/2022-06/2024

Program Leader of Planet X with Dr. James Miller, Professor of Humanities

Kunshan, China

- Led the team of Planet X game development with 6 members for 1 year, proposed and implemented a new flexible and decoupled program architecture, proposed a new GUI visual solution. Implemented a Python-based JSON generator for game content data management.
- Held regular meetings with team members enhancing communication and collaboration, coordinated progress and workflow throughout the development process.

Mathematical Contest in Modeling 2024 C – Honorable Mention	02/2024
Programmer & Modeling	Kunshan, China
• Proposed Psychological Angular Momentum Model (PAMM), a machine learning method to evaluate athletes' in-game performance and momentum. Integrated physical dynamics with time series model and applied various of method including Linear Regression, MLP and time series filtering. Skillfully integrated the modules including Pytorch, Pandas, Numpy, Scikit-learn. • XDzzzzzZyq/MCM-2024-C: MCM-2024 Problem C	

LEADERSHIP & SERVICE

Physics Journal Club	04/2024-Now
Seminar Presenter	Kunshan, China
• Delivered academic seminars on Physics-based Rendering (PBR) with demonstrations of my renderer, and on Computational Fluid Dynamics (CFD) covering Navier–Stokes fundamentals, numerical methods, and data-driven approaches.	
DKU Kendo Club	09/2023-12/2024
President	Kunshan, China
• Supported 6 Club Expos, took an active part in several club activities. Successfully held the first Kendo competition at DKU. Acquired the ability of communication and collaboration with members, faculties and coaches, and the ability to coordinate funds and nearly a hundred personnel in a comprehensive manner. • Awarded the rank of 1st kyu in the art of Kendo.	
DKU Innovation and Entrepreneurship	08/2022-12/2023
Design Manager	Kunshan, China
• Designed more than 30 posters, decoratives, and souvenirs. Supported the implementation of the Visual Identity System for DKU Innovation Center. Assisted with the on-site affairs of more than 3 large-scale events. Developed an automatic and user-friendly certificate-generating pipeline.	

TEACHING

CS101 Teaching Assistant	2025 Fall
• Led weekly office hours and lab sessions to support student learning. Guided students in programming fundamentals, debugging, and problem-solving. • Included 12 hours per month of coaching and instruction for tens of students	Kunshan, China
MATH105 Teaching Assistant	2025 Fall
• Led weekly office hours and recitation sessions to support student learning, reinforcing mathematical foundations, conceptual understanding, and rigorous problem-solving skills.	Kunshan, China

ADDITIONAL INFORMATION

Languages: Chinese Mandarin (Native), English (Proficient), Japanese (Beginner)
Skills: Python (Pytorch), C++, CUDA, OpenGL, Java/C#, Blender, Unity, After Effects, Photoshop.
Interests: Computer Graphics/Vision, Machine Learning/Deep Learning, Computational Physics/Biology.